Sin => deg -> val
asin => val -> rad

Pythagoras theorem :-

$x^2 + h^2 = c^2 \;/\; y^2 + h^2 = c^2$

$x^2 = c^2 - h^2$

$x = \sqrt{c^2 - h^2} \;/\; y = \sqrt{c^2 - h^2}$

$y^2 + h^2 = b^2 \;/\; x^2 + h^2 = b^2$

$y^2 = b^2 - h^2$

$y = \sqrt{b^2 - h^2} \;/\; x = \sqrt{b^2 - h^2}$

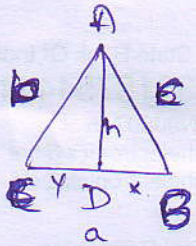
$\therefore\; a = x + y$

$$\boxed{\sqrt{c^2 - h^2} + \sqrt{b^2 - h^2} = a}$$

$\sin B = \frac{h}{c}$ $\qquad$ $\sin C = \frac{h}{b}$

$h = c \sin B$ $\qquad$ $h = b \sin C$

$h = c \sin B = b \sin C.$



$x = \sqrt{b^2 - h^2}$

$\therefore\; c^2 = h^2 + (a - x)^2$

$c^2 = h^2 + (a - \sqrt{b^2 - h^2})^2$

$c^2 = h^2 + (a^2 - 2a\sqrt{b^2 - h^2} + (b^2 - h^2)) \;/\; c^2 = \cancel{h^2} + a^2 + b^2 - \cancel{h^2} - 2a\sqrt{b^2 - h^2}$

$c^2 = a^2 + b^2 - 2a\sqrt{b^2 - h^2}$

Add "h^2" both sides

simplify

$h^2 = b^2 + \left[(c^2 - a^2 - b^2)/2a\right]^2$

$$-2a\sqrt{b^2-h^2} = c^2-a^2-b^2$$

$$\sqrt{b^2-h^2} = \frac{c^2-a^2-b^2}{-2a}$$

$$b^2-h^2 = \left[\frac{c^2-a^2-b^2}{2a}\right]^2$$

$$h^2 = b^2 - \left[\frac{c^2-a^2-b^2}{2a}\right]^2$$

$$h^2 = \frac{4b^2a^2 - \left[c^2-a^2-b^2\right]^2}{4a^2}$$

$$h^2 = \frac{(2ab-c^2+a^2+b^2)(2ab+c^2-a^2-b^2)}{4a^2}$$

$$h^2 = \frac{\left((a+b)^2-c^2\right)\left(c^2-(a-b)^2\right)}{4a^2}$$

$$h^2 = \frac{(a+b+c)(a+b-c)(c-a+b)(c+a-b)}{4a^2}$$

if $s = \frac{a+b+c}{2}$

$$h^2 = \cancel{16}^{4} \times s \times (s-c) \times (s-a) \times (s-b) \Big/ \cancel{4}a^2$$

$$h = 2 \times \sqrt{s(s-c)\times(s-a)\times(s-b)} \Big/ a$$

Substitute "h"

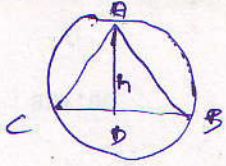
$$b \sin C = 2\sqrt{s(s-a)(s-b)(s-c)} \Big/ a$$

Both side $\div c$

$$\frac{\sin C}{c} = 2\sqrt{s(s-a)(s-b)(s-c)} \Big/ abc$$

$$\therefore \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = D \rightarrow \text{constant Diameter}$$

$$(\text{or})\ 2R$$



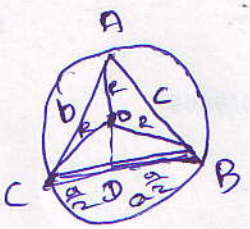
$$\frac{\cancel{c}}{2R} \times \frac{1}{\cancel{c}} = 2\sqrt{s(s-a)(s-b)(s-c)} \Big/ abc$$

$$R = abc \Big/ 4\,\text{root}\left(s.(s-a)(s-b)(s-c)\right)$$

$$\Delta(\text{delta}) = \sqrt{s.(s-a)(s-b)(s-c)}$$

$$\frac{a}{\sin A} = 2R$$

$$\sin A = \frac{a}{2R}$$



$$\sin A = \frac{\frac{a}{2}}{R}$$

$$\sin A = \frac{a}{2R}$$

$$\text{(by)}\ \sin B = \frac{b}{2R}$$

$$\sin C = \frac{c}{2R}$$

If you use asin

asin → acute
(-90°-+90°) negative
never obtuse

manully flip = 180°-acute
to obtuse

acos → can be ~~acute~~ acute
(0°-180°) be right angle
be obtuse